

Question 1A State representation.

[2 marks]

The state can be represented with five lists, 4 of the lists l_1 , l_2 , l_3 and l_4 will contain the sticks used for each side of the square, while the final list l_r will contain the *currently* unused sticks. Each list will contain the integers corresponding to the lengths of the sticks used for each side.

Question 1B Invariant.

[2 marks]

The total length of the sticks in each list l_1 , l_2 , l_3 and l_4 should not exceed one-fourth of the sum of all the stick lengths, represented as $\frac{1}{4} \sum_{i=0}^n s_i$. The union of all the lists is equivalent to the original list of sticks $[s_0, s_1, \dots, s_n]$.

Question 1C Initial and goal state.

[2 marks]

Initial State: Five lists: $l_1=[]$, $l_2=[]$, $l_3=[]$, $l_4=[]$, $l_r=[s_0, s_1, \dots, s_n]$.

Goal State: Where $T = \sum_i^n s_i$, $\text{sum}(l_1) = \text{sum}(l_2) = \text{sum}(l_3) = \text{sum}(l_4) = \frac{T}{4}$. $l_r = []$.

Question 1D Actions.

[2 marks]

In each step, the first edge from l_r is moved into one of l_1 , l_2 , l_3 and l_4 .

Question 1E Best search algorithm.

[4 marks]

We choose DFS because it will find a solution if one exists (i.e. it is complete) and constant space usage. We avoid BFS due to its higher space requirements. Similarly, IDS, while functional, incurs unnecessary overhead. DLS is not suitable as we do not prema facie know the correct depth limit.

[+1] each for brief explanations for each algorithm.

Note that the correct answer here depends on depends on Parts (A) to (D).

Question 1F Use Graph-Search or not?

[3 marks]

The answer depends on the details of the problem formulation. If all stick lengths are distinct ($s_i \neq s_j$ for all i, j where $i \neq j$), then there are no repeated states, and there is no necessity to employ GRAPH-SEARCH. However, if the box contains sticks with equal lengths ($s_i = s_j$ where $i \neq j$), this can lead to the generation of repeated states. In such cases, utilizing GRAPH-SEARCH allows us to avoid expanding the same states. However, to implement GRAPH-SEARCH efficiently, we will either have to ensure that the lists are sorted. This advantage comes at the cost of increased memory and time required to maintain a visited state set.

[+2] No need to use GRAPH-SEARCH and valid reason.

Question 1G Initial candidate solution.

[2 marks]

A possible candidate solution is a random allocation of the sticks into 4 lists representing the 4 sides $[s_0, s_1...]$, $[s_2, s_3...]$, $[s_4, s_5...]$, $[s_6, s_7...s_n]$. Each stick is allocated to one of the lists.

Question 1H Transition function.

[2 marks]

The transition function involves moving a stick from one list to another. For instance, if the current state is $\{[s_0, s_1\dots], [s_2, s_3\dots], [s_4, s_5\dots], [s_6, s_7\dots, s_n]\}$, a neighboring state could be $\{[s_1\dots], [s_0, s_2, s_3\dots], [s_4, s_5\dots], [s_6, s_7\dots, s_n]\}$.

If a student suggests swapping 2 sticks as the transition function, it is incorrect because we cannot be sure that swapping sticks along will necessarily allow the state representation to transition to the goal state, even if such a state exists, depending on the initial configuration.

Question 1I Heuristic function.

[2 marks]

We define:

$$h = \sum_{i \in \{0,1,2,3\}} \delta(l_i)$$

where $\delta(l_i)$ is an indicator function such that $\delta(l_i) = 1$ when l_i is equal to l_{avg} , and $\delta(l_i) = 0$ otherwise. Here, l_{avg} represents one-quarter of the total length of all the sticks provided by Ben Bitdiddle.

Upon reaching the goal state, we achieve $h = 4$, which is greater than any non-goal state.

Question 1J New formulation for Variant #1.

[6 marks]

However, as implied by Part (K) below, formulating the problem carefully so that the search will complete requires some thought. The most straightforward way to do so is to add a new list l_{unused} . In each step, instead of moving a stick from l_r into l_1, l_2, l_3 and l_4 , we can also move it into l_{unused} .

-2 marks for formulation that is correct, but that cannot guarantee termination.

Question 1K Best search algorithm for Variant #1.

[2 marks]

BFS. To minimize the number of sticks used, it is essential to arrange them in descending order, starting from the longest and ending with the shortest.

IDS with sticks arranged in descending order is also a viable choice since it enables us to find the optimal solution. Compared to BFS, it saves memory but node in the upper level of search tree are generated multiple times.

DFS is not recommended because it doesn't guarantee finding the minimal solution.

DLS is likewise not recommended for the same reason; it doesn't guarantee a minimal solution.

Question 1L Is your TREE-SEARCH complete and/or optimal? Is the search tree finite for Variant #1?

[3 marks]

If the answer to Part (K) is just BFS or IDS, then TREE-SEARCH is complete but not optimal.

For example, sticks = [1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2] and $n=2$

BFS solution: [1, 1], [1, 1], [1, 1], [1, 1] depth=8

Optimal solution: [2], [2], [2], [2] depth=12 (discards the first 8 sticks)

If the answer to Part (K) is BFS or IDS with sticks arranged in descending order, then TREE-SEARCH is both complete and optimal.

If the answer for Part (K) is DFS then the TREE-SEARCH is complete but not optimal.

If the answer for Part (K) is DLS then the TREE-SEARCH is not complete and not optimal.

Whether TREE-SEARCH will always terminate will depend on whether the search is set up carefully such that each stick in the box is used at most once. If the same stick can be used multiple times then the search is not guaranteed to terminate.

Question 2A Is h_A admissible? Explain.

[3 marks]

☐ Admissible ☒ Not admissible

Consider a scenario where the snake's current position is $\{(4, 2), (4, 3)\}$, and the goal position is $\{(4, 3), (4, 4)\}$. In this case, $h_A = 2$, which is greater than the optimal cost, $h^* = 1$.

Question 2B Is h_B admissible? Explain.

[3 marks]

☐ Admissible ☒ Not admissible

Consider a situation where the snake's current position is $\{(4,3), (3,3)\}$, and the goal position is $\{(4,3), (4,4)\}$. In this case, $h_B = 2$, which exceeds the optimal cost, $h^* = 1$.

Question 2C Is h_C admissible? Explain.

[3 marks]

☒ Admissible ☐ Not admissible

Admissible. The tail can move at most 1 step towards the goal tail in every move. If we consider a relaxed version of the Snake game where the only objective is to move the tail to its goal position, the real cost for this relaxed problem is indeed h_C . Therefore, h_C is admissible.

Question 2D Is h_D admissible and/or consistent? Explain.

[6 marks]

☐ Admissible ☒ Not admissible
☐ Consistent ☒ Not consistent

Not admissible. Consider the case where the current position of snake is $\{(4,3), (3,3)\}$ and goal is $\{(4,3), (4,4)\}$. Position of apple is $(4,4)$. In this case, $h_D = 1 + 1 > h^* = 1$.

Not consistent. After the snake eats the apple, all distances between the apple and other positions are reset to zero. Thus h_D holds a value of 0 for the goal. According to the above scenario (snake current position: $\{(4,3), (3,3)\}$; snake goal position: $\{(4,3), (4,4)\}$; apple position: $(4,4)$), the next node can be the goal node by rotating 90° . $h_D = 1 + 1 > c(n, a, n') + h_D(n') = 1 + 0$.

Question 2E Is h_E admissible and/or consistent? Explain.

[6 marks]

- ☒ Admissible ☐ Not admissible
☐ Consistent ☒ Not consistent

Admissible. $MD(h, ap)$ can decrease by a maximum of 2 for each move. In order to eat apple, we want to achieve $MD(h, ap) = 0$. The value $MD(h, ap)/2$ is an underestimate of the actual number of steps needed to move from the current head position to the apple. The snake's head must also advance towards the goal head position, and the same reasoning applies to $MD(ap, gh)$. Consequently, we can conclude that the heuristic function h_E is admissible.

Not consistent. Consider the case where the current position of snake is $\{(4,0), (4,1)\}$ and goal is $\{(4,3), (4,4)\}$. Position of apple is $(4,2)$. In this case, for the current node n , $h_E(n) = (1+2)/2$. By moving 1 cell to the right, a successor n' of n can be generated. The position of snake becomes $\{(4,1), (4,2)\}$ and the apple can be eaten, then all distances between the apple and other positions are reset to zero ($h_E(n') = 0$). Thus $h_E(n) = 1.5 > c(n, a, n') + h_E(n') = 1 + 0$.

Question 2F Choosing between h_E and h_F ? Explain.

[5 marks]

Consider the case where the current position of snake is $\{(4,3), (4,2)\}$ and goal is $\{(4,3), (4,4)\}$. Position of apple is $(4,4)$. In this case, $h_G = 2 + 0 > h^* = 1$.

Since h_F is not admissible, so we choose h_E .

Question 3A The eligibility of a low income loan taker?

[2 marks]

- ☐ Yes ☐ No ☒ Undefined

This question tests the student's ability to reason about a decision tree.

According to the given decision tree, we can't make a decision solely from the income.

Question 3B What is the entropy?

[2 marks]

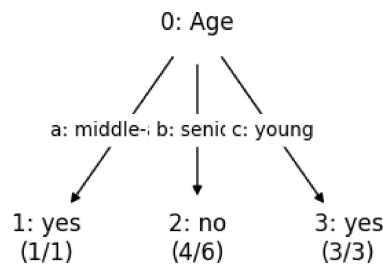
☐ $0.0 \leq \text{Entropy} < 0.2$ ☐ $0.2 \leq \text{Entropy} < 0.4$ ☐ $0.4 \leq \text{Entropy} < 0.6$ ☐ $0.6 \leq \text{Entropy} < 0.8$ ☒ $0.8 \leq \text{Entropy} \leq 1.0$

Show your working below (if you wish):

$$\begin{aligned} \text{Entropy} &= I\left(\frac{6}{10}, \frac{4}{10}\right) \\ &= -\frac{6}{10} \log_2\left(\frac{6}{10}\right) - \frac{4}{10} \log_2\left(\frac{4}{10}\right) \\ &= 0.971 \end{aligned}$$

Question 3C One-level Tree.

[6 marks]



Note: (X/Y) in the leaf nodes represent the number of samples belonging to the majority class over the total number of samples across all classes. To illustrate, a leaf node labeled as "no (4/6)" indicates that out of 6 samples, 4 of them belong to the "no" class.

What is node 0?

- | | | | |
|--------------------------------------|-------------------------------------|---------------------------------------|------------------------------|
| ☒ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☐ - |

What is node 1?

- | | | | |
|---------------------------|--------------------------------------|---------------------------------------|------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☒ yes | ☐ no/yes | ☐ - |

What is node 2?

- | | | | |
|-------------------------------------|-------------------------------------|---------------------------------------|------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☒ no | ☐ yes | ☐ no/yes | ☐ - |

What is node 3?

- | | | | |
|---------------------------|--------------------------------------|---------------------------------------|------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☒ yes | ☐ no/yes | ☐ - |

What is edge a?

- | | | | | | |
|---------------------------------|------------------------------|----------------------------|-----------------------------|--|--------------------------|
| ☐ excellent | ☐ fair | ☐ high | ☐ low | ☒ middle-aged | ☐ no |
| ☐ poor | ☐ senior | ☐ yes | ☐ young | ☐ - | |

What is edge b?

- | | | | | | |
|---------------------------------|---|----------------------------|-----------------------------|-----------------------------------|--------------------------|
| ☐ excellent | ☐ fair | ☐ high | ☐ low | ☐ middle-aged | ☐ no |
| ☐ poor | ☒ senior | ☐ yes | ☐ young | ☐ - | |

What is edge c?

- | | | | | | |
|---------------------------------|------------------------------|----------------------------|--|-----------------------------------|--------------------------|
| ☐ excellent | ☐ fair | ☐ high | ☐ low | ☐ middle-aged | ☐ no |
| ☐ poor | ☐ senior | ☐ yes | ☒ young | ☐ - | |

Question 3C One-level Tree. (Continued)

[6 marks]

Show your working below (if you wish):

$$\begin{aligned}
 \text{Remainder}(\text{Age}) &= \frac{1}{10} \left(I\left(\frac{1}{1}\right) \right) + \frac{6}{10} \left(I\left(\frac{4}{6}, \frac{2}{6}\right) \right) + \frac{3}{10} \left(I\left(\frac{3}{3}\right) \right) \\
 &= \frac{1}{10} (0.000) + \frac{6}{10} (0.918) + \frac{3}{10} (0.000) \\
 &= 0.000 + 0.551 + 0.000 \\
 &= 0.551
 \end{aligned}$$

$$\begin{aligned}
 \text{Remainder}(\text{CreditRating}) &= \frac{5}{10} \left(I\left(\frac{4}{5}, \frac{1}{5}\right) \right) + \frac{3}{10} \left(I\left(\frac{2}{3}, \frac{1}{3}\right) \right) + \frac{2}{10} \left(I\left(\frac{2}{2}\right) \right) \\
 &= \frac{5}{10} (0.722) + \frac{3}{10} (0.918) + \frac{2}{10} (0.000) \\
 &= 0.361 + 0.275 + 0.000 \\
 &= 0.636
 \end{aligned}$$

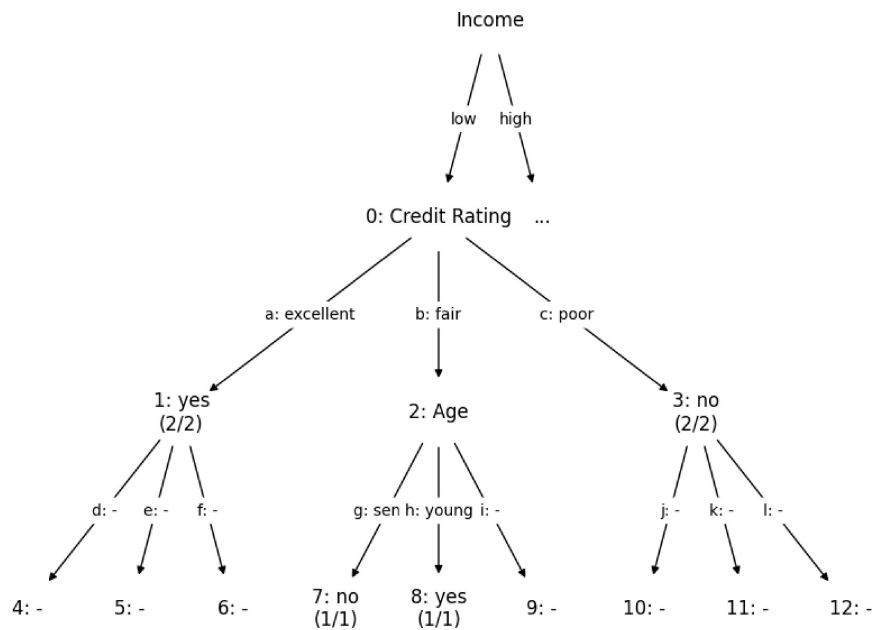
$$\begin{aligned}
 \text{Remainder}(\text{Income}) &= \frac{6}{10} \left(I\left(\frac{3}{6}, \frac{3}{6}\right) \right) + \frac{4}{10} \left(I\left(\frac{1}{4}, \frac{3}{4}\right) \right) \\
 &= \frac{6}{10} (1.000) + \frac{4}{10} (0.811) \\
 &= 0.600 + 0.325 \\
 &= 0.925
 \end{aligned}$$

$$\begin{aligned}
 \text{Remainder}(\text{CriminalRecord}) &= \frac{4}{10} \left(I\left(\frac{3}{4}, \frac{1}{4}\right) \right) + \frac{6}{10} \left(I\left(\frac{3}{6}, \frac{3}{6}\right) \right) \\
 &= \frac{4}{10} (0.811) + \frac{6}{10} (1.000) \\
 &= 0.325 + 0.600 \\
 &= 0.925
 \end{aligned}$$

We split based on "Age" as it results in the smallest remaining entropy.

Question 3D Full Decision Tree.

[6 marks]



What is node 0?

- ☐ Age ☒ Credit Rating ☐ Criminal Record ☐ no
☐ yes ☐ no/yes ☐ -

What is node 1?

- ☐ Age ☐ Credit Rating ☐ Criminal Record ☐ no
☒ yes ☐ no/yes ☐ -

What is node 2?

- ☒ Age ☐ Credit Rating ☐ Criminal Record ☐ no
☐ yes ☐ no/yes ☐ -

What is node 3?

- ☐ Age ☐ Credit Rating ☐ Criminal Record ☒ no
☐ yes ☐ no/yes ☐ -

What is node 4?

- ☐ Age ☐ Credit Rating ☐ Criminal Record ☐ no
☐ yes ☐ no/yes ☒ -

What is node 5?

- ☐ Age ☐ Credit Rating ☐ Criminal Record ☐ no
☐ yes ☐ no/yes ☒ -

What is node 6?

- ☐ Age ☐ Credit Rating ☐ Criminal Record ☐ no
☐ yes ☐ no/yes ☒ -

Are there any nodes from 7 onwards?

- ☒ Yes ☐ No

Question 3D Full Decision Tree. (Continued)

[6 marks]

What is edge a?

- | | | | | |
|--|----------------------------|-----------------------------------|--------------------------|----------------------------|
| ☒ excellent | ☐ fair | ☐ middle-aged | ☐ no | ☐ poor |
| ☐ senior | ☐ yes | ☐ young | ☐ - | |

What is edge b?

- | | | | | |
|---------------------------------|---------------------------------------|-----------------------------------|--------------------------|----------------------------|
| ☐ excellent | ☒ fair | ☐ middle-aged | ☐ no | ☐ poor |
| ☐ senior | ☐ yes | ☐ young | ☐ - | |

What is edge c?

- | | | | | |
|---------------------------------|----------------------------|-----------------------------------|--------------------------|---------------------------------------|
| ☐ excellent | ☐ fair | ☐ middle-aged | ☐ no | ☒ poor |
| ☐ senior | ☐ yes | ☐ young | ☐ - | |

What is edge d?

- | | | | | |
|---------------------------------|----------------------------|-----------------------------------|------------------------------------|----------------------------|
| ☐ excellent | ☐ fair | ☐ middle-aged | ☐ no | ☐ poor |
| ☐ senior | ☐ yes | ☐ young | ☒ - | |

What is edge e?

- | | | | | |
|---------------------------------|----------------------------|-----------------------------------|------------------------------------|----------------------------|
| ☐ excellent | ☐ fair | ☐ middle-aged | ☐ no | ☐ poor |
| ☐ senior | ☐ yes | ☐ young | ☒ - | |

What is edge f?

- | | | | | |
|---------------------------------|----------------------------|-----------------------------------|------------------------------------|----------------------------|
| ☐ excellent | ☐ fair | ☐ middle-aged | ☐ no | ☐ poor |
| ☐ senior | ☐ yes | ☐ young | ☒ - | |

Question 3D Full Decision Tree. (Continued)

[6 marks]

Show your working below (if you wish):

$$\begin{aligned}
 \text{Remainder}(\text{CreditRating}) &= \frac{2}{6} \left(I\left(\frac{2}{2}\right) \right) + \frac{2}{6} \left(I\left(\frac{1}{2}, \frac{1}{2}\right) \right) + \frac{2}{6} \left(I\left(\frac{2}{2}\right) \right) \\
 &= \frac{2}{6} (0.000) + \frac{2}{6} (1.000) + \frac{2}{6} (0.000) \\
 &= 0.000 + 0.333 + 0.000 \\
 &= 0.333
 \end{aligned}$$

$$\begin{aligned}
 \text{Remainder}(\text{Age}) &= \frac{1}{6} \left(I\left(\frac{1}{1}\right) \right) + \frac{4}{6} \left(I\left(\frac{1}{4}, \frac{3}{4}\right) \right) + \frac{1}{6} \left(I\left(\frac{1}{1}\right) \right) \\
 &= \frac{1}{6} (0.000) + \frac{4}{6} (0.811) + \frac{1}{6} (0.000) \\
 &= 0.000 + 0.541 + 0.000 \\
 &= 0.541
 \end{aligned}$$

$$\begin{aligned}
 \text{Remainder}(\text{CriminalRecord}) &= \frac{3}{6} \left(I\left(\frac{2}{3}, \frac{1}{3}\right) \right) + \frac{3}{6} \left(I\left(\frac{2}{3}, \frac{1}{3}\right) \right) \\
 &= \frac{3}{6} (0.918) + \frac{3}{6} (0.918) \\
 &= 0.459 + 0.459 \\
 &= 0.918
 \end{aligned}$$

We split based on "Credit Rating" as it results in the smallest remaining entropy.

If Credit Rating = "fair", then we split again:

	Age	Criminal Record	Loan
4	senior	no	no
9	young	yes	yes

Based on the priority order, we should split based on "Age".

Question 3E Precision, Recall, F1.

[5 marks]

Number of True Positives (TP)?

- ☐ 0 ☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ >4

Number of False Positives (FP)?

- ☒ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ >4

Number of True Negatives (TN)?

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ >4

Number of False Negatives (FN)?

- ☐ 0 ☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ >4

Precision?

- ☐ $0.0 \leq \text{Precision} < 0.2$ ☐ $0.2 \leq \text{Precision} < 0.4$
☐ $0.4 \leq \text{Precision} < 0.6$ ☐ $0.6 \leq \text{Precision} < 0.8$
☒ $0.8 \leq \text{Precision} \leq 1.0$

Recall?

- ☐ $0.0 \leq \text{Recall} < 0.2$ ☐ $0.2 \leq \text{Recall} < 0.4$
☒ $0.4 \leq \text{Recall} < 0.6$ ☐ $0.6 \leq \text{Recall} < 0.8$
☐ $0.8 \leq \text{Recall} \leq 1.0$

F1 Score?

- ☐ $0.0 \leq \text{F1 Score} < 0.2$ ☐ $0.2 \leq \text{F1 Score} < 0.4$
☐ $0.4 \leq \text{F1 Score} < 0.6$ ☒ $0.6 \leq \text{F1 Score} < 0.8$
☐ $0.8 \leq \text{F1 Score} \leq 1.0$

Show your working below (if you wish):

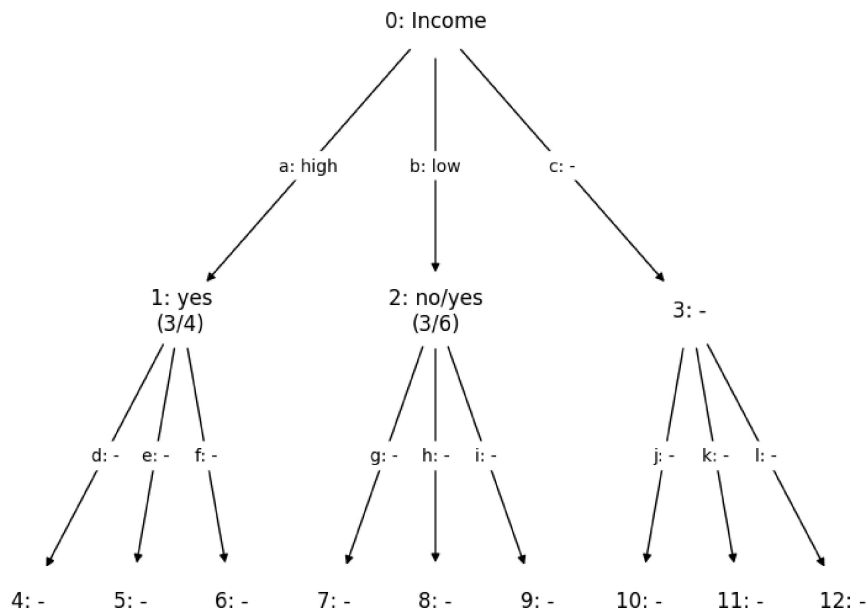
$$\begin{aligned}
 \text{Precision} &= \frac{TP}{TP + FP} \\
 &= \frac{3}{3 + 0} \\
 &= 1.000
 \end{aligned}$$

$$\begin{aligned}
 \text{Recall} &= \frac{TP}{TP + FN} \\
 &= \frac{3}{3 + 3} \\
 &= 0.500
 \end{aligned}$$

$$\begin{aligned}
 F1 &= \frac{2}{\frac{1}{P} + \frac{1}{R}} \\
 &= \frac{2}{\frac{1}{1.000} + \frac{1}{0.500}} \\
 &= 0.667
 \end{aligned}$$

Question 3F Min-Sample DT.

[3 marks]



What is node 0?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|---|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☒ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☐ - |

What is node 1?

- | | | | |
|---------------------------|--------------------------------------|---------------------------------------|------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☒ yes | ☐ no/yes | ☐ - |

What is node 2?

- | | | | |
|---------------------------|-------------------------------------|---|------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☒ no/yes | ☐ - |

What is node 3?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|------------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☒ - |

What is node 4?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|------------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☒ - |

What is node 5?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|------------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☒ - |

What is node 6?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|------------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☒ - |

Are there any nodes from 7 onwards?

- | | |
|---------------------------|-------------------------------------|
| ☐ Yes | ☒ No |
|---------------------------|-------------------------------------|

Question 3F Min-Sample DT. (Continued)

[3 marks]

What is edge a?

- ☐ excellent ☐ fair ☒ high ☐ low ☐ middle-aged ☐ no
- ☐ poor ☐ senior ☐ yes ☐ young ☐ -

What is edge b?

- ☐ excellent ☐ fair ☐ high ☒ low ☐ middle-aged ☐ no
- ☐ poor ☐ senior ☐ yes ☐ young ☐ -

What is edge c?

- ☐ excellent ☐ fair ☐ high ☐ low ☐ middle-aged ☐ no
- ☐ poor ☐ senior ☐ yes ☐ young ☒ -

What is edge d?

- ☐ excellent ☐ fair ☐ high ☐ low ☐ middle-aged ☐ no
- ☐ poor ☐ senior ☐ yes ☐ young ☒ -

What is edge e?

- ☐ excellent ☐ fair ☐ high ☐ low ☐ middle-aged ☐ no
- ☐ poor ☐ senior ☐ yes ☐ young ☒ -

What is edge f?

- ☐ excellent ☐ fair ☐ high ☐ low ☐ middle-aged ☐ no
- ☐ poor ☐ senior ☐ yes ☐ young ☒ -

Show your working below (if you wish):

Question 4A Game Tree.

[6 marks]

What is the minimum number of variables required to maintain the game state?

- ☐ 1
 ☐ 2
 ☒ 3
 ☐ 7
 ☐ 8
 ☐ Others

How many nodes (not including the leaves) are there in the game tree?

- ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7
☐ 8 ☒ 9 ☐ 10 ☐ 11 ☐ 12 ☐ 13 ☐ 14
☐ 15 ☐ 16 ☐ 17 ☐ 18 ☐ 19 ☐ 20 ☐ >20

What is the number of leaves where the first player wins?

- ☐ 0 ☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ >6

What is the number of leaves where the first player loses?

- ☐ 0 ☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ >6

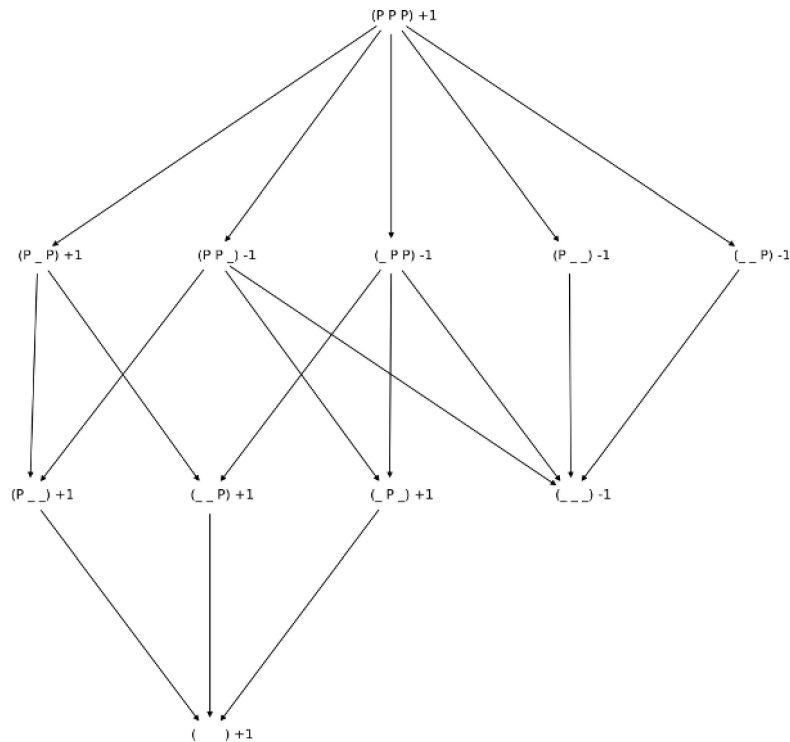
What is the total number of edges in the game tree?

- ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7
☐ 8 ☐ 9 ☐ 10 ☐ 11 ☐ 12 ☐ 13 ☐ 14
☐ 15 ☐ 16 ☐ 17 ☒ 18 ☐ 19 ☐ 20 ☐ >20

Which player do we expect to win?

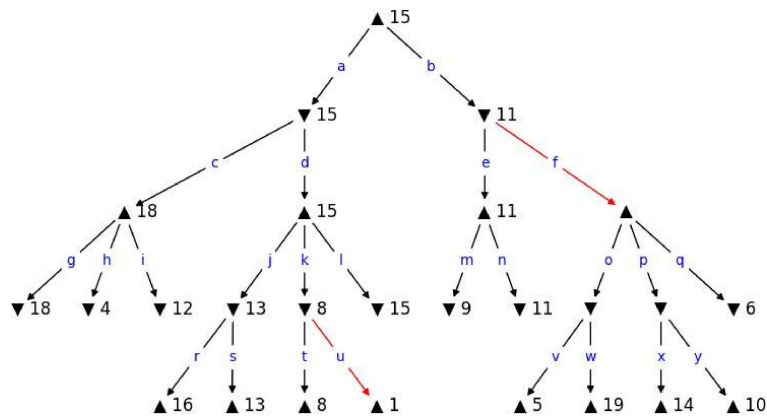
- ☒ First player
 ☐ Second player
 ☐ It depends
 ☐ Draw

Show your working below (if you wish):



Question 4B Alpha-beta left-to-right.

[5 marks]



What is the root value?

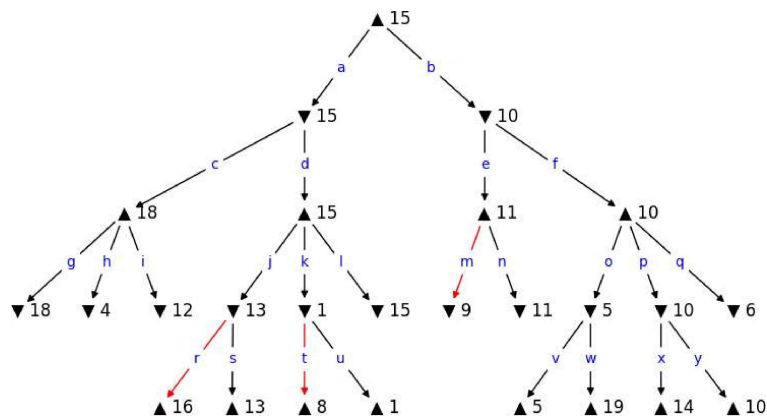
- ☐ 1 ☐ 4 ☐ 5 ☐ 6 ☐ 8 ☐ 9 ☐ 10 ☐ 11
☐ 12 ☐ 13 ☐ 14 ☒ 15 ☐ 16 ☐ 18 ☐ 19

Which of the following link(s) are pruned? Shade all that is/are true.

- ☐ a ☐ b ☐ c ☐ d ☐ e ☒ f ☐ g
☐ h ☐ i ☐ j ☐ k ☐ l ☐ m ☐ n
☐ o ☐ p ☐ q ☐ r ☐ s ☐ t ☒ u
☐ v ☐ w ☐ x ☐ y

Question 4C Alpha-beta right-to-left.

[4 marks]

Which of the following link(s) are pruned? Shade all that is/are true.

- ☐ a ☐ b ☐ c ☐ d ☐ e ☐ f ☐ g
☐ h ☐ i ☐ j ☐ k ☐ l ☒ m ☐ n
☐ o ☐ p ☐ q ☒ r ☐ s ☒ t ☐ u
☐ v ☐ w ☐ x ☐ y

Question 5 3 most important lessons

[3 marks]

There are no right answers here. Students will get credit for 3 well-explained learning points that are reasonable and justified. Student needs to put in *some* effort. Clearly if the student makes a patently false statement, marks will be deducted.